

Test 1: showChunk with a file in a subdirectory

Source file is `code/stats.R`. The chunk output must land in `generated/code/stats.R-chunkXX.txt`, requiring `generated/code/` to be created automatically.

```
cat("n_=", n, "\n")
```

```
cat("mean_=", round(mean(x), 3), "\n")
```

```
cat("sd_=", round(sd(x), 3), "\n")
```

Test 2: showChunk with a flat file (no subdirectory)

Source file is `flatcode.R` in the current directory. Output lands in `generated/flatcode.R-flat_hello.txt`.

```
cat("hello_from_flat_file\n")
```

```
cat("2_+2_=", 2 + 2, "\n")
```

Test 3: Stream exhaustion — 18 showChunk calls

Calling `showChunk` 18 times on the same file exercises the pre-allocated stream fix. The old code would crash after the 16th call with “No room for a new read stream.”

```
cat("min_=", round(min(x), 3), "\n")
```

```
cat("max_=", round(max(x), 3), "\n")
```

```
cat("median_=", round(median(x), 3), "\n")
```

```
q <- quantile(x, c(0.25, 0.75))
cat("Q1_=", round(q[1], 3), "\n")
```

```
cat("Q3_=", round(q[2], 3), "\n")
```

```
cat("IQR_=", round(IQR(x), 3), "\n")
```

```
cat("var_=", round(var(x), 3), "\n")
```

```
cat("range_=", round(diff(range(x)), 3), "\n")
```

```
cat("sum_=", round(sum(x), 3), "\n")
```

```
cat("skewness_proxy_=", round(mean((x - mean(x))^3) / sd(x)^3, 3), "\n")
```

```

cat("kurtosis_proxy=", round(mean((x - mean(x))^4) / sd(x)^4, 3), "\n")

cat("cv=", round(sd(x) / mean(x), 3), "\n")

cat("se=", round(sd(x) / sqrt(n), 3), "\n")

ci <- mean(x) + c(-1, 1) * 1.96 * sd(x) / sqrt(n)
cat("95%CI: [", round(ci[1], 3), ",", round(ci[2], 3), "]\n")

cat("All done.\n")

```

Test 4: runRChunk with a subdirectory file

runRChunk should extract the chunk, run it via the R server, and display the output.

```
Error in eval(ei, envir) : object 'n' not found
```

```
Error in eval(ei, envir) : object 'x' not found
```

```
Error in eval(ei, envir) : object 'x' not found
```

Test 5: runRChunk with a flat file

```
hello from flat file
```

```
2 + 2 = 4
```

Test 6: Missing chunk label (should show red error, not crash)

Label *no_such_label* not found in code/stats.R

Test 7: Missing source file (should show red error, not crash)

Source file *no_such_file.R* not found